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Quantitative interpretation of divergence between PM_{10} and $PM_{2.5}$ mass measurement by TEOM and gravimetric (Partisol) instruments

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Abstract

An intercomparison of PM_{10} and $PM_{2.5}$ mass measured simultaneously by tapered element oscillating microbalance (TEOM) instruments and Partisol samplers, with laboratory gravimetric determination of particle mass from the latter, has been carried out at a rural site in Oxfordshire, UK. The TEOM gives lower readings of both PM_{10} and $PM_{2.5}$ mass and the underestimation is greater at high concentrations. The strong day-to-day variation between instruments is the result of the variability of the particle composition and the influence of temperature and relative humidity. Results show that most of the particulate material not measured by the TEOMs belongs to the $PM_{2.5}$ fraction and suggest that the loss of semi-volatile compounds constitutes an important part of the difference between instruments; even though it is not the only cause of divergence. Ammonium nitrate is one of the major particulate compounds lost in the inlet of the TEOMs located at Harwell. Its volatilisation can explain a significant part of the difference between TEOMs and filter-based Partisol methods. At Harwell, events of high concentrations are often events of high concentrations of semi-volatile compounds including particulate nitrate and as a consequence, the TEOM instrument may miss events above the daily air quality standard of $50\,\mu g\,m^{-3}$. The influences of temperature and relative humidity are demonstrated. The difference between measurements decreases with increasing temperature and with decreasing relative humidity. This result is consistent with the mass deficit being associated with loss of semi-volatile substances such as NH_4NO_3 whose formation is temperature and humidity-dependent. It may also be related to positive artefacts associated with the filter-based method such as particle-bound water and adsorbed semi-volatile components.
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1. Introduction

Under European Directive 1999/30/CE, limit values for PM_{10} have been set. These require a daily average concentration of $50\,\mu g\,m^{-3}$ with 35 annual exceedences permitted, and a maximum annual average of $40\,\mu g\,m^{-3}$. These concentrations should be measured by the European gravimetric reference method or a method shown as equivalent. Member states that cannot fulfil this obligation have to establish a relationship between their method and the reference method necessary to convert data for the purposes of reporting under the above-mentioned directive.

PM_{10} concentrations have been measured using tapered element oscillating microbalance (TEOM)

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instruments for many years at more than 350 sites in the UK, including the 50 urban background, kerbside, industrial and rural sites belonging to the Automatic Urban and Rural Network (AURN). The TEOM instrument has the advantage over conventional gravimetric methods of particle mass monitoring to provide data on an almost real-time basis and to be a cost and labour-effective method. Their introduction in the UK has led to significant improvements in knowledge of the sources and behaviour of particulate matter (QUARG, 1996; APEG, 1999). In comparison, 24-h measurements with filter-based gravimetric methods are costly and time consuming. The filter handling involves a large number of steps including pre-conditioning, weighing of blanks, filter installation and filter removal on the sampling site, post-conditioning and weighing of dust-loaded filters.

However, many studies comparing various filter-based PM_{10} (or $PM_{2.5}$) samplers with TEOM instruments have shown that TEOMs report lower particle mass values than the collocated filter-based samplers (Allen et al., 1997; Ayers et al., 1999; Soutar et al., 1999; APEG, 1999; Salter and Parsons, 1999; Williams and Bruckmann, 2001; Cyrys et al., 2001). This is attributed to heating of the inlet of the TEOM system, usually to 50°C, conducted initially in order to minimise interferences from the evaporation and condensation of water onto the filter and to provide a stable and reproducible measurement (Patashnick and Rupprecht, 1991). The particulate material lost in the inlet of the TEOM is thought to be mainly semi-volatile particulate matter and particle-bound water, both of which partition between condensed and vapour phases in the atmosphere. The major such components are expected to be ammonium nitrate and semi-volatile organic compounds.

It is important to highlight that filtration-based mass measurements are not artefact-free. They can also lead to significant volatilisation losses of the same semi-volatile components during collection, due to changes in temperature, relative humidity, composition of the aerosol or pressure drop across the filter, and after collection—due to filter handling, transport and storage (see Chow, 1995). Artefacts in the measurement of particle mass concentrations from gravimetric methods also arise from the adsorption of semi-volatile organic gases onto or from collected particulate matter and filter media (Turpin et al., 2000) and the neutralisation of acid or basic gases on either filter media, or collected particulate matter (Tsai and Huang, 1995). Artefacts may also be associated with particle-bound water that may constitute a significant part of particulate matter. The variability of the water content of particles and the hysteresis in the water adsorption–desorption pathways of some major atmospheric compounds (Seinfeld and Pandis, 1996) complicate the mass measurement. In recent works, Speer et al. (2003) show that water associated with both inorganic and hygroscopic organic components of $PM_{2.5}$ can contribute significantly to mass and Price et al. (2003) reveal that particle-bound water may be the major cause of difference between TEOM and reference gravimetric measurements.

The amounts of semi-volatile compounds and particle-bound water vary both temporally and spatially. As a consequence, the extent of the difference between TEOMs and filter-based gravimetric methods is not universal, and spatial and seasonal differences have been shown by many studies (Allen et al., 1997; APEG, 1999; Williams and Bruckmann, 2001) implying that the relationship between TEOMs and filter-based gravimetric methods should be examined for each site. This also implies that the relationship between TEOMs and gravimetric methods may not be proportional or linear and may be complex.

This paper describes a study of the relationship between PM_{10} and $PM_{2.5}$ mass measurements by TEOM and Partisol (gravimetric) instruments at a rural site, and an investigation of the causal factors.

2. Material and methods

2.1. Description of the site

The monitoring station is within a self-contained, air-conditioned building located in the Harwell Research Science Centre in Oxfordshire. The nearest road is for access to buildings within the Science Park only. The manifold inlet is approximately 3 m above ground level. The surrounding area is generally open with agricultural fields. The nearest trees are at a distance of 200–300 m from the monitoring station. The station is a rural site that may be influenced by the London urban plume and possibly the busy A34 highway at a distance of 2 km to the east.

2.2. Description of instruments and gravimetric measurements

2.2.1. Tapered element oscillating microbalance

Two 1400A series TEOMs (Rupprecht and Patashnick Co., Inc.) have been providing measurements of PM_{10} and $PM_{2.5}$ at Harwell on an almost real-time basis for a number of years. The particle mass is determined by continuous weighing of particles deposited onto a filter. The filter is attached to a vibrating hollow tapered glass tube. The frequency of mechanical oscillation of this tube is a function of its mass. Deposition of particles on the filter leads to changes in the mass of the system and results in changes of its frequency of oscillation. A microprocessor directly converts the vibration frequency to mass concentrations.

The flow rate through the analyser is controlled using thermal mass flow controllers and is automatically

measured to determine the mass concentration. Air at $16.67\,l\,min^{-1}$ ($1.002\,m^3\,h^{-1}$) is sampled through the sampling head and divided between the filter flow ($3\,l\,min^{-1}$) and an auxiliary flow ($13.67\,l\,min^{-1}$). The inlet is heated to 50°C prior to particles being deposited onto the filter in order to eliminate the effect of condensation or evaporation of particle water. An identical impactor is used as a size-selective inlet for PM_{10} on both instruments. For $PM_{2.5}$, the Partisol and the TEOM after 23 January 2002 were fitted with a Sharp-Cut Cyclone inlet; prior to that date the TEOM was fitted with a URG $PM_{2.5}$ cyclone inlet.

2.2.2. Partisol Plus Model 2025

The Partisol Plus Model 2025 from Rupprecht and Patashnick Co., Inc. is a microprocessor controlled measuring device for the sampling of particulate matter. Two Partisol instruments were installed on the Harwell site in order to measure simultaneously the PM_{10} and $PM_{2.5}$ particles on a daily basis. These instruments use the same size-selective inlet designs as the ones used on the TEOM with the same flow rate of $16.67\,l\,min^{-1}$ ($1.002\,m^3\,h^{-1}$). In contrast to the TEOM instrument, the inlet is not heated and air is sampled at ambient temperature and pressure.

The Partisol has a fully automatic filter exchange mechanism that provides unattended monitoring for up to 16 consecutive days for 24-h sampling prior to filter change. Temperature and pressure sensors are provided in the instrument with internal regulators to maintain the temperature within ± 5°C of ambient temperature. The total volume of air sampled, total measuring time, average temperature and pressure are recorded by the microprocessor. The Partisol Plus Model 2025 is equivalent to the gravimetric reference method according to the European standard prEN 12341 for the collection of PM_{10} (see Mückler, 2000).

Quartz fibre filters (Whatman QMA 47 mm diameter filters, 0.6 μm pore size) have been used for the collection of particulate matter. Pre-conditioning and post-conditioning of filters were undertaken in accordance with the requirements of prEN 12341. Blank and dust-loaded filters are handled according to the same protocols. Filters are equilibrated for 48 h within an air-conditioned weighing room at a temperature of 20°C and a relative humidity of 50% before weighing on a balance with a resolution of 10 μg.

2.3. Data for inclusion in the comparison

The TEOM concentrations as reported by the instrument include a built-in calibration factor of 1.03∗'TEOM reading' + 3 μg. This calibration factor has been determined through intercomparison of data from TEOMs and filter-based reference instruments sampling a standard Arizona road dust in order to achieve US EPA certification (Patashnick and Rupprecht, 1991). The values corrected by the US EPA calibration factor correspond to data as supplied by the UK network (AURN data).

Examination of the influence of this calibration factor on the relationships between TEOM and Partisol PM_{10} data for seven sites in the UK has shown that the calibration factor applied to TEOM values explains a large component of intercepts significantly higher than zero (results not shown). These intercepts have no physical meaning. On the contrary, linear models computed with raw TEOM readings have intercepts close to zero (for the relationship of Partisol versus TEOM with $R^2 > 0.70$). In this paper, the raw TEOM readings (i.e. with the calibration factor removed) have been used because they correspond to the true mass collected with the TEOM instrument.

PM_{10} and $PM_{2.5}$ have been measured at the Harwell site using two collocated TEOMs for a number of years. Gravimetric data from two Partisol Plus 2025 have been derived from samples from September 2000. Data from 30 September 2000 to 2 July 2002 for the PM_{10} and from 7 September 2000 to 19 June 2002 for the $PM_{2.5}$ are included in the comparison. A total of 436 paired 24-h PM_{10} data and 461 paired 24-h $PM_{2.5}$ data are available for this study.

Particulate nitrate data used in this paper have been derived from samples collected using a collocated Partisol Plus Model 2025 instrument with PTFE filters and measured by ion chromatography.

Meteorological data (temperature and dew point used to compute relative humidity data) have been provided by the British Atmospheric Data Centre (BADC).

3. Results

3.1. Comparison of TEOM with gravimetric Partisol particle mass

Table 1 presents the mean and median PM_{10} and $PM_{2.5}$ particle mass concentrations collected with the filter-based Partisol samplers and TEOMs. Standard deviations and interquartile ranges are computed to give an estimation of how the particle mass concentrations are spread out. Robust estimators (median and interquartile range) are also presented because they are not influenced by occasional high concentrations like the mean and the standard deviation.

In agreement with other studies (Allen et al., 1997; Ayers et al., 1999; Soutar et al., 1999; APEG, 1999; Salter and Parsons, 1999; Williams and Bruckmann, 2001; Cyrys et al., 2001), the results show that the TEOMs give lower particle mass than the filter-based gravimetric methods. The median relative differences

Table 1
Average PM_{10} and $PM_{2.5}$ concentrations and number of exceedence days for PM_{10} for the period September 2000–July 2002 ($N = 436$ for PM_{10}; $N = 461$ for $PM_{2\cdot5}$)

	PM concentrations		Ratios Partisol/TEOM
	Partisol	TEOM[a]	
PM_{10}			
Arithmetic mean ± SD	18.1 (±10.2)	10.2 (±5.3)	1.8 (±0.51)
Median ± IQR[b]	15.7 (±9.9)	9.2 (±5.5)	
Daily concentration $> 50\,\mu g/m^3$	8	0	
$PM_{2.5}$			
Arithmetic mean ± SD	12.3 (±9.9)	6.5 (±4.2)	2.1 (±2.7)
Median ± IQR	9.0 (±8.8)	5.3 (±4.4)	

[a] TEOM data are raw mass with USEPA correction removed.
[b] IQR = interquartile range.

between Partisol and TEOM particle mass concentrations are, respectively, 42% for PM_{10} and 48% for $PM_{2.5}$.

The median PM_{10} and $PM_{2.5}$ concentrations show smaller differences between Partisol and TEOM data than the means, indicating that some high concentrations measured by the Partisol instruments contribute to the higher difference in the means. The higher standard deviations (or interquartile ranges) for Partisol data than for TEOM data lead to the same conclusion. This means that the TEOMs read lower to a greater extent at high concentrations and suggests that higher amounts of semi-volatile compounds characterise events of high particulate matter concentrations.

The mean ratio Partisol/TEOM and the corresponding standard deviation for the whole studied period are presented in Table 1. Ratios of 1.8 for PM_{10} data and of 2.1 for $PM_{2.5}$ data are found, both with high standard deviations, showing that a large range of ratios occurs. This result is in agreement with other studies (King et al., 2000; Cyrys et al., 2001; Green et al., 2001) showing the variability of the difference between measurements from TEOMs and gravimetric methods from one day to another.

The European Limit Value for particles measured as PM_{10} is $50\,\mu g\,m^{-3}$ (24 h mean) with up to 35 permitted exceedences per year. The number of days exceeding this daily standard for the September 2000–July 2002 period is also presented in Table 1. No exceedence days were measured by the TEOM, while the gravimetric Partisol method gives a total of 8 days exceeding $50\,\mu g\,m^{-3}$. Most of these events corresponded to daily Partisol concentrations above $60\,\mu g\,m^{-3}$, while the corresponding TEOM particle mass concentrations were generally below $30\,\mu g\,m^{-3}$. Seven of these events corresponded to cold days (temperature below 10°C) including four to very cold days (temperature below 5°C) and three of these events corresponded to days with high relative humidity (above 90%). This suggests that semi-volatile compounds and occasionally particle-bound water on Partisol filters are responsible for large differences between instruments. For all these events, the $PM_{2.5}$ fraction represented more than 80% of the PM_{10} fraction (and often more than 90%); while the average $PM_{2.5}/PM_{10}$ ratio for the studied period is 60%. This means that episodes of high concentrations of PM_{10} concentrations at Harwell are driven by high $PM_{2.5}$ concentrations. The mean ratio Partisol/TEOM for these events of high concentrations is 2.1, which is higher than the mean ratio for the whole study period, confirming the greater difference at high concentrations.

The best fitting curves for the TEOM-Partisol relationships are linear despite the higher differences at high concentrations (see Fig. 1). Some studies have actually found nonlinear relationships due to higher amounts of volatile species at higher particle mass concentrations; resulting in higher divergences at higher concentrations (Salter and Parsons, 1999; APEG, 1999). Despite the scatter of the data, reasonable linear regressions between TEOMs and Partisol data can be established (Fig. 1). These linear regressions have been computed using the reduced major axis (RMA) regression method. Unlike the least square regression method that assumes that observations of the independent (X) variable are accurate, this method makes no assumption about the set of X-variable observations. A recent paper from Ayers (2001) has shown that the least square regression analysis is not appropriate for an instrument comparison exercise and leads to biased evaluations of the relationship between two instruments.

The relationships found for Harwell are not improved using nonlinear models and not really improved nor changed by removing outlier points. These results show that a large part of Harwell particle mass collected with the Partisol is vaporized in the inlet of the TEOMs. Almost all TEOM particle mass concentrations are lower than the Partisol particle mass concentrations.

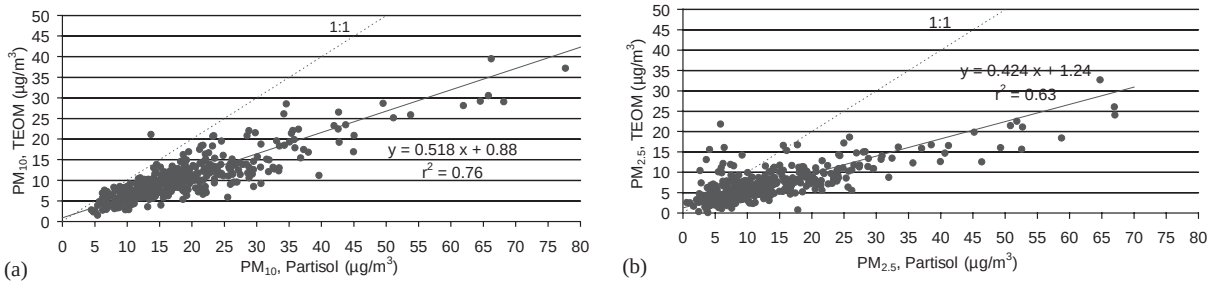


Fig. 1. TEOM versus Partisol for (a) PM_{10} and (b) $PM_{2.5}$ ($N = 436$ for PM_{10}; $N = 461$ for $PM_{2.5}$).

Possible explanations for these substantial underestimations are examined below.

3.2. Origins of the divergence and influencing parameters

The results suggest that some semi-volatile compounds are lost in the inlet of the TEOMs. The major inorganic semi-volatile compound is thought to be particulate ammonium nitrate. Its gas–particle partitioning depends on the atmospheric temperature and relative humidity (Stelson and Seinfeld, 1982a, b). Particulate ammonium chloride is another well-known semi-volatile inorganic compound (Pio and Harrison, 1987a, b). Except in specific polluted areas, the concentrations of ammonium chloride in the atmosphere are generally too low to lead to condensation, and particulate ammonium chloride is considered negligible in this study. A large range of semi-volatile organic compounds in small amounts but of non-negligible total mass are contained in particulate matter. The gas–particle partitioning of these organic semi-volatile compounds depends on a large number of parameters such as their concentration, the total particle surface area, the particle composition, the total particulate matter concentration, the atmospheric temperature and the relative humidity (US EPA, 1996). Semi-volatile compounds are expected to be mainly in the fine fraction of particles. Particulate ammonium nitrate and semi-volatile organic compounds are thought to be the main compounds lost in the inlet of the TEOM instrument.

Particle-bound water associated with particles collected with the Partisol constitutes another explanation for the difference between the two instruments. The equilibration of Partisol filters to a relative humidity of 50% does not remove all particle-bound water. The examination of the particle water content as a function of relative humidity shows a significant increase for humidity larger than 30% (US EPA, 1996). Due to hysteresis effects the mass of water is greater in samples collected at high humidity (as generally prevails in the UK) and transferred to a lower humidity environment. A recent study (Price et al., 2003) concluded that the retention of water by the filter of the gravimetric method

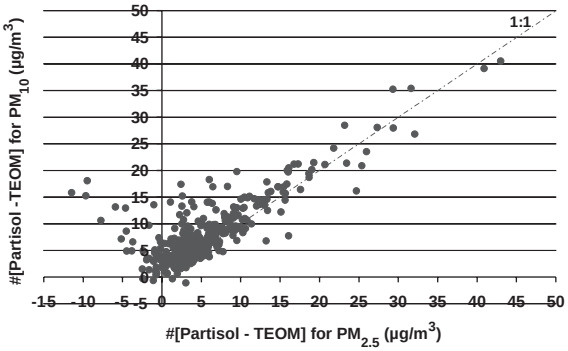


Fig. 2. Difference between Partisol and TEOM for PM_{10} data versus difference between Partisol and TEOM for $PM_{2.5}$ data ($N = 370$)—data in $\mu g\,m^{-3}$.

was the major cause of differences with the TEOM for their site in Northern England.

Finally, adsorption of organic vapour by the Quartz filters used in the Partisol instruments may contribute to the difference between the two instruments. Quartz fibre filters have a large specific surface area upon which adsorption of gases can occur. A large positive artefact due to adsorption of organic gases onto Quartz fibre filters is now well-known (Turpin et al., 2000). The importance of this artefact associated with the sampling of particulate matter with Quartz fibre filters cannot be estimated in this study. The contribution of particulate ammonium nitrate and the influence of temperature and relative humidity are examined, since appropriate data were available.

The close agreement between the difference between Partisol and TEOM data for PM_{10} and for $PM_{2.5}$ shows that most of the material not measured by the TEOMs is contained in the $PM_{2.5}$ fraction (see Fig. 2). The series of negative values of differences for $PM_{2.5}$ (i.e. TEOM data higher than Partisol data) are considered as suspicious. These values are not associated with negative values of differences for PM_{10}. They may be the result of artefacts with the $PM_{2.5}$ TEOM sampler. Allen et al. (1997) have observed large fluctuations in the $PM_{2.5}$ mass concentrations measured with the TEOM due to rapid changes

in the composition of air masses (pollutant concentrations and ambient relative humidity) affecting the equilibrium of particles collected on the TEOM filter and then responsible for adsorption or desorption of particulate material. They did not observe the same phenomenon for the simultaneous PM_{10} TEOM measurements and assumed that the coarse fraction of particles stabilises the semi-volatile particulate matter collected simultaneously. If these suspicious values are removed, a close linear relationship of $y = 0.973x + 2.167$, $r^2 = 0.78$ is found between the difference for PM_{10} and difference for $PM_{2.5}$. A slope close to 1 with a small systematic difference of about $2\,\mu g\,m^{-3}$ (intercept) is shown by this relationship. About the same difference is found comparing the average differences between Partisol and TEOM data for PM_{10} and $PM_{2.5}$ (median of differences for $PM_{2.5}$, $4.0\,\mu g\,m^{-3}$; for PM_{10}, $6.3\,\mu g\,m^{-3}$). This suggests that a small part of the difference between the two methods is in general associated with the PM_{coarse} fraction. Nevertheless, a few much larger differences between Partisol and TEOM data for PM_{10} than for $PM_{2.5}$ are observed for low values of differences for $PM_{2.5}$. These much larger differences (above $15\,\mu g\,m^{-3}$) occurred during damper days (relative humidity above 90%) and might be the result of particle-bound water associated with PM_{coarse} particles (e.g. a major compound, NaCl is very hygroscopic and is mainly found in the coarse fraction of particulate matter).

The particulate ammonium nitrate mass was computed from the particulate nitrate assuming that all the particulate nitrate is associated with ammonium ions. In practice involatile nitrates (e.g. sodium nitrate) may make an important contribution to airborne nitrates. The particulate ammonium nitrate is compared to the difference between the two instruments and also used to correct the TEOM particle mass concentrations. Fig. 3 represents the relationships between the sums of TEOM particle mass data and calculated particulate ammonium nitrate with Partisol particle mass data; and in Fig. 4, the relationships between the difference between Partisol and TEOM data with the calculated particulate ammonium nitrate are plotted.

Both PM_{10} and $PM_{2.5}$ TEOM data are significantly improved by adding the particulate ammonium nitrate. The slopes of the relationships with the Partisol data are closer to the 1:1 line and the correlation coefficients are better showing that the data are less scattered. For the

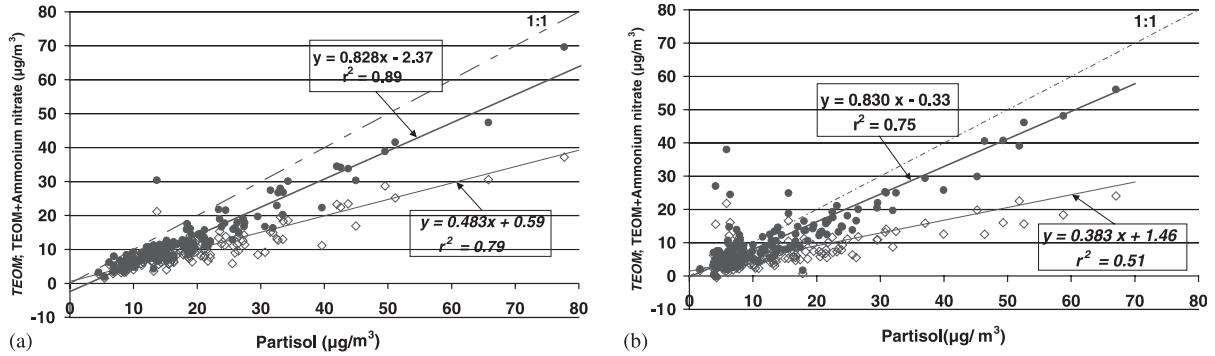


Fig. 3. TEOM PM data (empty dots) and TEOM PM data + particulate ammonium nitrate (full dots) versus Partisol PM data: (a) PM_{10} ($N = 146$) and (b) $PM_{2.5}$ ($N = 161$).

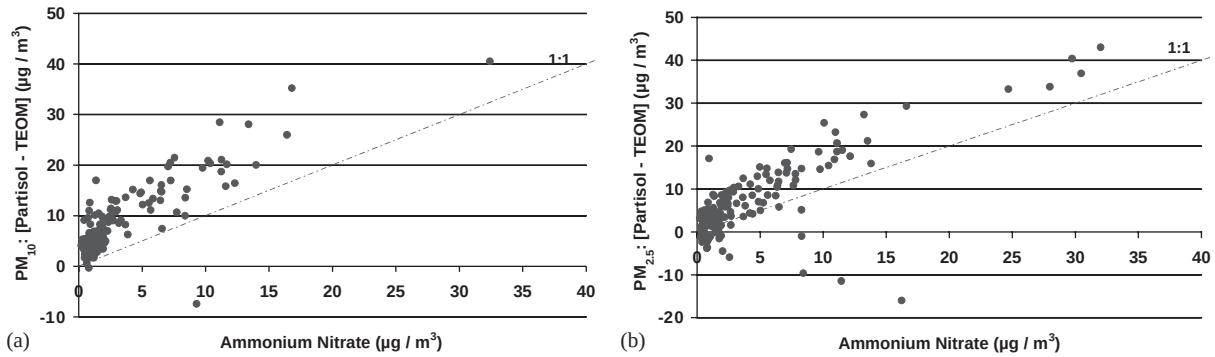


Fig. 4. Difference between Partisol and TEOM PM data versus calculated ammonium nitrate: (a) PM_{10} ($N = 146$) and (b) $PM_{2.5}$ ($N = 161$)—data in $\mu g\,m^{-3}$.

concentrations of particulate ammonium nitrate higher than 2–3 $\mu g\,m^{-3}$, the relationships between the difference between Partisol and TEOM data and the particulate ammonium nitrate are fairly linear (see Fig. 4); while for the concentrations lower than 2–3 $\mu g\,m^{-3}$, the relationship is poor indicating a significant relative contribution of other species. The remaining differences between Partisol and TEOM data are on average about 2.7 and 4.9 $\mu g\,m^{-3}$, respectively, for $PM_{2.5}$ and PM_{10} and seem to be constant for the higher differences. Particulate ammonium nitrate (median concentration about 1.9 $\mu g\,m^{-3}$, during the study period) is lost in the inlet of the TEOMs and contributes substantially to the lower reading of the TEOM, but its volatilisation does not explain the whole difference between Partisol and TEOM data. It corresponded on average to about 26% of the material lost in the PM_{10} fraction and about 40% of the material lost in the $PM_{2.5}$ fraction at Harwell. Different results in other studies reflect the different aerosol composition at different sites and explain the wide range of relationships between TEOM and gravimetric reference method found in the literature. Allen et al. (1997) found that the entire difference between a TEOM and a manual filter-based method for the Rubidoux site (California) can be attributed to particulate ammonium nitrate. On the contrary, Cyrys et al. (2001) have a small contribution of ammonium nitrate at their site. Harwell lies between these two extremes.

At the Harwell site, particulate ammonium nitrate represents an important part of the particulate material not measured with the TEOM instruments during episodes of high particulate pollution. Particulate nitrate data are available for three out of the eight episodes for which the gravimetric PM_{10} particle mass exceeded 50 $\mu g\,m^{-3}$. These three events were all associated with a high concentration of particulate nitrate (equivalent NH_4NO_3 concentrations above 16 $\mu g\,m^{-3}$) and the particulate ammonium nitrate represented more than 50% of the particulate material lost. For two of them, the TEOM data are successfully corrected by adding the particulate ammonium nitrate; while for the last one, the loss of the particulate ammonium nitrate does not explain the entire difference between the two instruments (about 18% is still uncorrected). The five other episodes with $PM_{10} > 50\,\mu g\,m^{-3}$ measured with the gravimetric Partisol method and not measured with the TEOM have ratios Partisol/TEOM above the mean ratio Partisol/TEOM and occurred during cold days, indicating that high amounts of semi-volatile compounds were lost. These results suggest that episodes dominated by high secondary particulate pollutants are not measured by the TEOM at the Harwell site.

The amount of semi-volatile compounds associated with the particles is expected to depend on the temperature and the relative humidity. Additionally, the relative humidity has a significant impact on the growth of particulate matter. Comparisons between TEOMs and reference gravimetric methods in different countries have shown that for warmer and dryer regions the agreement is better than for colder and damper regions (Williams and Bruckmann, 2001; Noack et al., 2001). Similarly, other studies have shown that the agreement is better during the warmer months of the year than during the colder months (Allen et al., 1997; Williams and Bruckmann, 2001). Price et al. (2003) reveal that the divergence at their sampling site is higher on damper and/or rainy days. All these studies seem to confirm the influence of these meteorological parameters, but a previous study (Cyrys et al., 2001) did not find any relationship between the underestimation of the TEOM and the temperature or the relative humidity.

Because of the seasonal variation of the underestimation of the TEOM, Williams and Bruckmann (2001) recommend in their "Guidance to member states on PM monitoring and intercomparisons with the reference method" examination of the seasonal variations of factors and equations to amend the TEOM data. This examination for the Harwell site is not very helpful because of the small contrast between the seasons (ratio Partisol/TEOM for summer, 1.73; for winter, 1.88).

The relative differences (differences between the gravimetric Partisol mass and the TEOM mass divided by the Partisol mass) versus the temperature and versus the relative humidity are presented, respectively, in Figs. 5a and b. These results confirm that the differences between the TEOM and the Partisol for PM_{10} mass concentrations measured at Harwell depend on both the temperature and the relative humidity. The difference between the two instruments decreases with the temperature and increases with relative humidity. However, a large range of relative differences is associated with each temperature and relative humidity bin, suggesting a simultaneous influence of the particle composition. The same results are found for the $PM_{2.5}$ at Harwell and for other sites in the UK (results not shown).

Fig. 5a clearly shows that the differences between Partisol and TEOM data are higher than 40% for lower temperatures (below 2°C) and lower than 25% for higher temperatures (above 18°C). Fig. 5b shows that half of the differences between Partisol and TEOM data are below 30% for relative humidity lower than 70% and higher than 50% for relative humidity higher than 90%. The examination of the ratio Partisol/TEOM with both temperature and relative humidity leads to similar conclusions. The mean ratio for colder and damper weather ($T < 10°C$ and $RH > 80\%$) is 1.98 and for warmer and drier weather ($T > 10°C$ and $RH < 80\%$) is 1.63.

The decrease of the difference between the TEOM and the Partisol with temperature is likely to be related to a decrease in the amount of semi-volatile compounds in

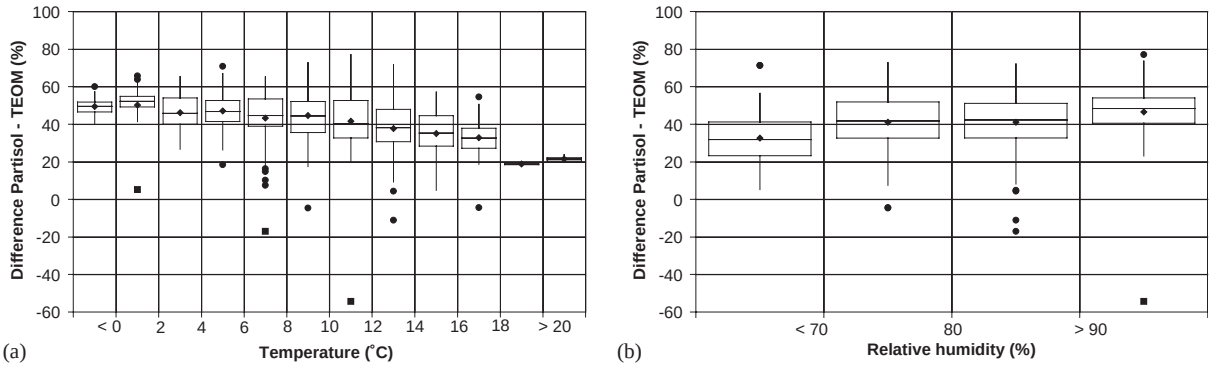


Fig. 5. Box plots of relative differences between Partisol and TEOM PM_{10} data for different (a) temperature ranges and (b) relative humidity ranges.

the particles or to an increase of the semi-volatile material volatilised from the Partisol filters, while the increase with relative humidity is likely to be related to the increase of the water content of particles and possibly also to the increase of semi-volatile compounds in particulate matter (as an example, the equilibrium of the ammonium nitrate between gaseous and condensed phases depends on the relative humidity, see Stelson and Seinfeld, 1982a, b). The influence of relative humidity on the difference between TEOM and gravimetric data seems to be less important than the influence of the temperature. The larger range of percentages associated with each range of humidity might be the result of the variation of particle composition. It might also be the result of the hysteresis of some major particulate compounds (e.g. $(NH_4)_2SO_4$) responsible for complicated relationships between particles and water (Speer et al., 2003). The significant influence of the temperature suggests that the lower reading of the TEOM is mainly due to the loss of semi-volatile compounds in the TEOM inlet.

4. Conclusion

An intercomparison between two TEOMs and two Partisol samplers for the monitoring of particles as PM_{10} and $PM_{2.5}$ has been carried out at a rural site located in Oxfordshire, UK. Results confirm the expected large difference between TEOM reading and particle mass from filter-based gravimetric methods. On average almost half of the particle mass derived from the Partisol instruments is not measured by the TEOMs. Because most of the particulate material not measured by the TEOMs is in the $PM_{2.5}$ fraction of particulate matter, the relative difference between the two methods is larger for $PM_{2.5}$ than for PM_{10}. Strong day-to-day variations of the difference between TEOM and Partisol particle mass are observed. This variability leads to difficulty in establishing statistical relationships that successfully amend TEOM data.

Results suggest that the major semi-volatile compound lost in the inlet of the TEOMs is ammonium nitrate. Adding the calculated ammonium nitrate mass to the TEOM mass significantly improves the mass estimate, accounting for about 40% of the $PM_{2.5}$ mass deficit. Loss of other particulate material such as semi-volatile organic compounds and water must also contribute since the loss of particulate ammonium nitrate does not explain the entire difference. Large differences between measurements are associated with high particulate ammonium nitrate concentrations. As a consequence, episodes of high particulate matter concentration dominated by secondary pollutants at Harwell are not measured by the TEOMs.

The influences of temperature and of relative humidity on the difference between TEOMs and Partisol particle mass have been demonstrated. Elevated temperatures or lower relative humidities reduce the divergence between measurements. This is in agreement with previous studies showing that in summer or in areas with warm and dry climates, the agreement between TEOMs and reference gravimetric methods is better than in winter or in areas with colder and damper climates. These influences are consistent with the material lost in the inlet of the TEOMs being semi-volatile compounds and particle-bound water. The loss of semi-volatile substances from the Partisol filters reducing the divergence between the two methods during elevated temperatures is not precluded.

The establishment of a model to correct the TEOM particle mass by including influential variables such as particle composition, temperature and relative humidity would be a better alternative than statistical models. However, this study suggests that semi-volatile organic compounds may constitute a significant part of the particulate material lost in the TEOMs. This component of particulate matter includes a large range of

compounds that cannot be measured on a daily basis and moreover, the physico-chemical properties of the vast majority of such compounds are poorly quantified. On the other hand, artefacts from filter-based gravimetric methods possibly contribute to the difference and should be quantified.

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